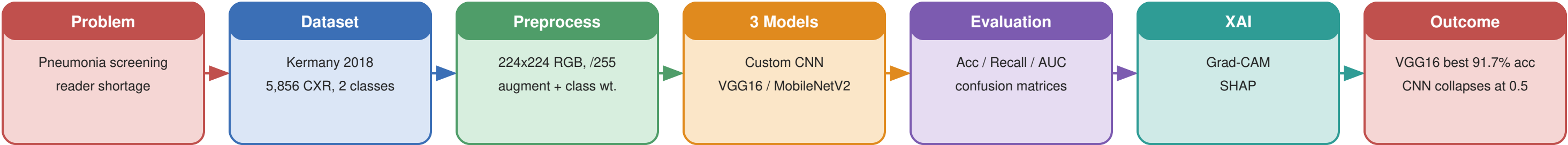


Explainable custom-CNN vs. transfer-learning framework for chest X-ray pneumonia detection



Key test-set results (624 images: 234 NORMAL / 390 PNEUMONIA)

Model	Accuracy	PNEU recall	NORMAL recall	AUC	Params
Custom CNN	0.625	1.000	0.000	0.914	0.52M
VGG16	0.917	0.974	0.821	0.969	14.85M
MobileNetV2	0.905	0.951	0.829	0.970	2.59M

Take-home: transfer learning (VGG16) gives the best balanced accuracy; the from-scratch CNN ranks well (AUC 0.914) but collapses at the 0.5 threshold, a calibration issue. Grad-CAM and SHAP confirm lung-field attention.